Mixing Matters: Evidence Position Bias Across Sequence Mixers in Long-Context Question Answering

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Abstract

Language models often use relevant evidence less effectively when it appears in the middle of a long context. We ask whether this position bias changes with the mechanism used to mix information across tokens. For each multi-document question, we move the answer-bearing document through ten positions while holding the question and distractors fixed. In a shared-corpus comparison near 2.8B parameters, Pythia shows a clear beginning advantage that two Mamba variants lack, while all three models show an end advantage. A tighter comparison between pure and hybrid Mamba-2 models points in the same direction, but its paired effect remains statistically uncertain. Across additional exploratory studies, the family gap appears only once the compared models can answer the task, a corpus swap changes overall accuracy without a detectable change in curve shape, and Pythia reproduces the beginning advantage on synthetic retrieval. Prompt placement can materially reshape both edges. Attention-sink measurements and position probes motivate hypotheses about evidence use, but do not establish a causal mechanism. All ten-document QA results use an exploratory split, and the held-out confirmatory split remains unexamined.

1 Introduction

Retrieval-augmented systems often present a model with several documents and expect it to use whichever one contains the answer. Yet evidence is not equally usable everywhere in the prompt: accuracy commonly falls when the relevant document moves from an edge toward the middle (Liu et al., 2024). This failure is important even when retrieval itself succeeds, because a downstream model can still neglect the retrieved evidence. Long-context benchmarks expose related weaknesses across question answering, summarization, and retrieval (Bai et al., 2024; Shaham et al., 2023; Li et al., 2024).

The usual lost-in-the-middle curve has mostly been studied in Transformers. Dense attention can revisit any earlier token directly, whereas state-space models such as Mamba compress the prefix into a recurrent state (Gu and Dao, 2024; Dao and Gu, 2024). This architectural difference could change how evidence position affects an answer. It is difficult to test cleanly, however, because model families also differ in training data, tokenizer, depth, scale, positional encoding, and capability.

We study the question with a paired intervention on evidence position. For every question, the same answer document and distractors form one bundle, and only the answer document's position changes. The main comparison finds a beginning advantage for Pythia that is absent in two similarly sized Mamba models trained on the same corpus, while the end advantage is shared. A more tightly matched pure-versus-hybrid Mamba-2 comparison is directionally compatible but does not detect

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a reliable paired attention effect. Scale, corpus, task, production-model, prompt, sink, and probe studies then define where the observation does and does not generalize.

The paper makes three contributions. First, it provides an auditable position-intervention harness with anchors, preserved outputs, paired inference, and an untouched confirmatory split. Second, it separates a strong exploratory family interaction from a tighter but statistically uncertain attention comparison. Third, it turns the remaining phases into bounded evidence: corpus and task checks test scope, while prompt interventions and internal measurements generate mechanistic hypotheses without being presented as causal proof.

2 Related work

Position bias in long contexts Liu et al. (2024) established the lost-in-the-middle curve in retrieval and multi-document QA. This task derives its questions from Natural Questions (Kwiatkowski et al., 2019), while LongBench, ZeroSCROLLS, and LooGLE broaden long-context evaluation across tasks, languages, and reasoning demands (Bai et al., 2024; Shaham et al., 2023; Li et al., 2024). Subsequent work measures effective context length (Hsieh et al., 2024b), associates the curve with attention allocation (Hsieh et al., 2024a), and proposes reordering, compression, retrieval, and context-use interventions (An et al., 2024; Peysakhovich and Lerer, 2023; Jiang et al., 2024; Xu et al., 2024; Agrawal et al., 2024). Irrelevant context can itself distract a model, motivating our use of the same distractors at every tested position (Shi et al., 2023). We focus on whether the curve changes across sequence mixers.

Sequence mixers Dense self-attention (Vaswani et al., 2017) mixes token pairs under a causal mask and commonly encodes order with rotary embeddings or linear attention biases (Su et al., 2024; Press et al., 2022). Alternative long-sequence mixers include structured state spaces, recurrent weighted key-value models, and long convolutions (Gu et al., 2022; Peng et al., 2023; Poli et al., 2023). Mamba and Mamba-2 carry information through selective state-space recurrences (Gu and Dao, 2024; Dao and Gu, 2024). Hybrid models such as Griffin, Jamba, and Nemotron-H combine recurrent or state-space blocks with attention (De et al., 2024; Lieber et al., 2024; NVIDIA, 2025). Waleffe et al. (2024) release the pure and hybrid 8B Mamba-2 checkpoints used in our tightest controlled comparison.

Cross-architecture long-context evidence Architecture comparisons do not imply that recurrent models are uniformly position-robust. Huang (2025) find practical long-context failures across recurrent, hybrid, and Transformer designs, and a concurrent preprint reports a U-shaped Mamba recall curve on structured memory tasks (Airlangga et al., 2025). These results differ from our Mamba QA curve and reinforce a central boundary of this study: position profiles depend on the task and prompt, not only the mixer. Work that reformulates Mamba recurrence as implicit attention offers a complementary route for future mechanism tests (Ali et al., 2025).

Attention sinks and probes Xiao et al. (2024) showed that causal transformers can concentrate attention on initial tokens, and Sun et al. (2024) connected unusually large internal activations to concentrated attention. We test whether this attention-sink statistic covaries with primacy, treating the result as correlational because a valid sink-blocking ablation is absent. Our position probe follows the principle that a diagnostic classifier needs a control task before its accuracy can be interpreted as evidence about a representation (Hewitt and Liang, 2019).

3 Measurement and statistics

Task and anchors We use the 2,655 released questions from the Lost-in-the-Middle 10-document QA setting (Liu et al., 2024), which is built from Natural Questions (Kwiatkowski et al., 2019). Each question has one gold document and nine distractors. We build ten prompts per question, moving only the gold document through positions 0 to 9 while retaining the same distractors and token budget. A no-gold floor measures guessing or memorized knowledge, and a gold-only ceiling measures answerability under perfect retrieval. The fixed split uses seed 240521 and contains 800 exploratory and 1,855 confirmatory questions. Every reported ten-document QA result uses the exploratory allocation or a subset of it. The confirmatory allocation remains unopened.

Edges and statistics Let a_S, a_M, and a_E be mean accuracy at start positions $\{0, 1\}$, middle positions $\{4, 5\}$, and end positions $\{8, 9\}$. We define primacy as $a_S - a_M$ and recency as $a_E - a_M$. The question is the unit of analysis. Each of 10,000 bootstrap samples draws complete ten-position question bundles with replacement, following the bootstrap principle of resampling the independent unit (Efron, 1979), and model comparisons reuse the same sampled questions. We report percentile 95% confidence intervals and apply Holm's sequential correction to the primacy and recency tests within each comparison family (Holm, 1979). The intervals quantify variation across questions under the fixed checkpoints, prompts, and decoding configuration, not variation across training seeds or model checkpoints.

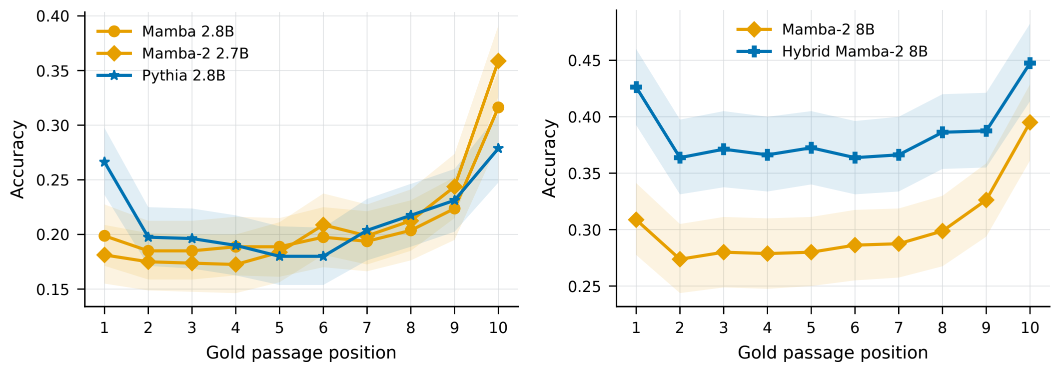
Reproducibility and controls The harness vendors the official prompt builder and `best_subspan_em` scorer at a pinned upstream commit. It also records normalized exact match, following the normalization convention popularized by SQuAD (Rajpurkar et al., 2016), and first-line extraction. Generation is greedy with temperature 0, at most 32 new tokens, fixed seeds, BF16 where supported, pinned model revisions, and a fixed Mamba execution path within each controlled contrast. Each phase fixes one execution path within its controlled comparison, although the Transformers and Megatron backends differ across phases. Committed standard QA sweeps preserve prompts, raw generations, failures, token counts, software versions, and decoding settings, except Phase 6, for which only summaries and plots are committed. The released key-value task provides an executed positive control: the pipeline detects a $+0.38$ Pythia edge-versus-middle effect. Negative and distractor-order controls are implemented but have no committed run artifacts, so we do not count them as passed controls.

4 Architecture comparisons

Primary 2.8B comparison We compare `pythia-2.8b` (Biderman et al., 2023) with `mamba-2.8b` (Gu and Dao, 2024) and `mamba2-2.7b` (Dao and Gu, 2024). All are base models pretrained on the Pile (Gao et al., 2020), have similar parameter counts, and use related tokenizers. Table 1 and Figure 1 show a positive primacy edge for Pythia and no positive primacy edge for either Mamba model. The paired Mamba-minus-Pythia primacy interactions are -0.053 and -0.070 (both Holm $p < 10^{-4}$), while Mamba-1 versus Mamba-2 is uncertain ($+0.017$, 95% CI $[-0.005, 0.038]$). All three models have positive recency edges. These comparisons match corpus and approximate scale, but not depth, positional encoding, state size, or checkpoint family.

Table 1: Pile-trained models on 800 exploratory questions. Intervals use a question-level paired bootstrap; p is Holm-corrected.

Model	Primacy (95% CI)	Recency (95% CI)	Floor / Ceiling
`pythia-2.8b`	$+0.052$ $[0.032, 0.072]$, $p<10^{-4}$	$+0.075$ $[0.054, 0.097]$, $p<10^{-4}$	$0.091/0.640$
`mamba-2.8b`	-0.001 $[-0.016, 0.014]$, $p=0.91$	$+0.077$ $[0.058, 0.097]$, $p<10^{-4}$	$0.115/0.615$
`mamba2-2.7b`	-0.018 $[-0.034, -0.003]$, $p=0.02$	$+0.105$ $[0.086, 0.124]$, $p<10^{-4}$	$0.128/0.619$



(a) Pile-trained models near 2.8B parameters **(b) Matched pure and hybrid Mamba-2 models at 8B**

Figure 1: Accuracy as the answer-bearing document moves through the context on 800 exploratory questions. Pythia rises at the beginning, while both smaller Mamba variants do not; the matched 8B hybrid also rises more than the pure model, but their paired difference remains uncertain. Shaded bands show 95% question-bootstrap confidence intervals.

Matched 8B attention comparison Our strongest control uses NVIDIA's pure and hybrid 8B Mamba-2 checkpoints (Waleffe et al., 2024), which share training data, tokenizer, scale, depth, and positional setup. The hybrid contains 7% attention layers. Within models, the pure model's primacy edge is uncertain ($+0.008$, $p = 0.36$) and the hybrid's is positive ($+0.027$, $p = 0.007$). One estimate being significant and the other not does not establish that the models differ. The paired hybrid-minus-pure primacy effect is $+0.019$ (95% CI $[-0.006, 0.044]$, Holm $p = 0.144$), while the recency effect is -0.028 (95% CI $[-0.055, -0.001]$, Holm $p = 0.089$). This direction supports the 2.8B observation but does not isolate a causal attention effect.

Table 2: Matched 8B primacy comparison on 800 exploratory questions. The paired interaction is the relevant architecture test.

Contrast	Primacy	95% CI	Holm p
pure Mamba-2	$+0.008$	$[-0.009, 0.024]$	0.362
hybrid	$+0.027$	$[0.008, 0.047]$	0.007
hybrid minus pure	$+0.019$	$[-0.006, 0.044]$	0.144

5 Scope checks

Scale Across five approximate size pairs, the Pythia-minus-Mamba primacy gap is near zero at the two smallest scales and is $+0.062$, $+0.069$, and $+0.053$ in the three larger pairs (Figure 3). The smallest Pythia ceiling is only 0.009, so this is a descriptive capability-and-family trend, not evidence for a depth threshold or a causal scale transition.

Training corpus Holding the Mamba-2.8B architecture fixed, the Pile-minus-SlimPajama primacy effect is -0.007 (95% CI $[-0.030, 0.016]$, $p = 0.568$) and the recency effect is -0.016 (95% CI $[-0.044, 0.012]$, Holm $p = 0.502$) (Soboleva et al., 2023). The floor differs substantially (0.270 versus 0.114), so this comparison finds no edge-shape change but does not rule out training-data effects (Figure 3).

Synthetic retrieval On 50 RULER `niah_single_1` examples at 2K tokens (Hsieh et al., 2024b), Pythia again has a primacy edge ($+0.12$, 95% CI $[0.05, 0.20]$). Both Mamba models score 1.0 at every depth, so this check reproduces the transformer pattern but cannot compare mixer curves on the synthetic task (Figure 4).

Descriptive system comparison We also run `Llama-3.1-8B` (Dubey et al., 2024), `Qwen2.5-7B` (Yang et al., 2024), and `Nemotron-H-8B` (NVIDIA, 2025). All three show positive primacy edges

($+0.076$, $+0.041$, $+0.067$). Only Llama versus Qwen has a significant paired primacy interaction ($+0.036$, Holm $p = 0.016$). Because the systems differ on architecture, data, tokenizer, token count, depth, and positional encoding, these results establish prevalence, not an architectural explanation (Figure 6).

6 Mechanistic evidence and intervention

Attention sinks track primacy Using the initial-token attention-share statistic of Xiao et al. (2024), we measure almost no final-layer token-0 sink in Pythia-160M (0.003), while Pythia-2.8B reaches 0.455. Pythia-410M has a small final-layer sink (0.034) and no significant primacy edge. Figure 5 shows a scale-linked association, but the measurements share model scale as a confound and no valid sink ablation was completed.

Utilization, not storage, is a hypothesis A balanced linear probe on frozen mid-depth states predicts edge versus middle position above its shuffled-label control for Pythia (0.603 versus 0.484) and Mamba (0.648 versus 0.508). Thus position information remains linearly decodable even when Mamba has no positive primacy edge. This supports a utilization hypothesis, but probe decodability does not prove that the model stores or uses the answer-bearing content.

Training-free prompt intervention On 200 exploratory questions, repeating the question both before and after the documents changes Mamba primacy from 0.000 to $+0.103$ ($p < 10^{-4}$). Placing the question only before the documents does not create primacy ($+0.020$, $p = 0.366$), but reduces recency from $+0.070$ to $+0.010$. The bookend result is not a mitigation result: much of the larger edge comes from lower middle-position accuracy. It also changes query repetition and token placement together, so it is not a clean test of fixed-state compression.

Prompt sensitivity Changing the prompt template reduces Pythia primacy from $+0.070$ to $+0.005$ in the 200-question mechanism subset. This result warns against treating the measured magnitude as a template-invariant architectural constant.

7 Discussion and limitations

The most secure result is the 2.8B exploratory family-by-position interaction. The matched 8B direction, sink association, and probe results motivate a mixer-dependent evidence-use hypothesis, while prompt sensitivity shows that architecture is not the only determinant.

Several limitations set the confirmatory boundary. All analyses use the exploratory split, so the planned 1,855-question test must be run under a frozen analysis before confirmatory language is warranted. The 2.8B and scale comparisons retain depth, positional-encoding, state-size, and capability confounds. The matched 8B paired contrast crosses zero after Holm correction, and its run manifest records a dirty source tree, weakening exact reproducibility. The negative and distractor-order controls lack committed outputs. A 50-generation blinded audit sample exists, but human labels were not completed. Finally, the attempted hybrid sink block did not alter the model because its custom attention path did not honor the hook; it is not an ablation result. The next decisive study should freeze exclusions and contrasts, complete these controls, repair the hybrid intervention, and evaluate the held-out split once.

Broader impact Position-sensitive evidence use can make retrieval-augmented systems unreliable even when retrieval succeeds. An evaluation that exposes this failure can support better ordering strategies and more honest model selection. The same phenomenon can also cause systems to privilege facts because of placement rather than relevance, so reported long-context accuracy should not be treated as uniform evidence access. This work releases diagnostics and outputs, not a new generative model.

8 Conclusion

On the exploratory Lost-in-the-Middle split, Pythia-2.8B and two Pile-trained Mamba models have different primacy profiles under a paired ten-position evaluation. A matched 8B comparison is directionally consistent but statistically uncertain, so the present evidence supports a family-by-position interaction without establishing that attention causes it. The held-out split and the remaining control work provide a clear path from this exploratory workshop result to a confirmatory study.

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A Supplementary evidence and artifacts

This appendix maps every experimental phase to its executed evidence and includes the most informative committed plots that do not fit in the main paper. All figures are generated from the cited JSON summaries by one deterministic plotting script. The canonical figure files are SVG; the PDF companions embedded here are produced from the same plotting calls because standard LaTeX engines do not include SVG directly.

A.1 Phase and artifact ledger

Table 3 separates executed measurements from their allowed interpretation. The standard ten-position sweep contains ten prompts per question and uses the same question bundle at every position.

Table 3: Executed evidence by phase. Sample counts are questions unless otherwise stated.

Phase	Executed comparison	Committed source	Interpretation in this paper
1	200 QA; 50 key-value	`artifacts/phase1`	Calibration and positive control
2	3 models, $n = 800$ each	`artifacts/phase2`	Primary family interaction
3	pure and hybrid 8B, $n = 800$	`artifacts/phase3`	Matched but uncertain contrast
4	5 size pairs, $n = 800$ each	`artifacts/phase4`	Descriptive scale trend
5	2 corpora, $n = 800$ each	`artifacts/phase5`	Corpus scope check
6	3 models, 50 at two lengths	`artifacts/phase6`	Synthetic task scope check
7	100-200 per intervention; 1,200 states	`artifacts/phase7-mechanisms`	Hypothesis generation only
8	3 systems, $n = 800$ each	`artifacts/phase8`	Descriptive prevalence check

The dataset snapshot contains 2,655 rows and has SHA-256 checksum `192a05b27af2b09eec33ca0c94bb5cf82bcaf70d78b3bdff1258df34bf37aab9`. The fixed split allocates 800 rows to exploration and 1,855 to confirmation. The confirmatory rows have not been evaluated.

A.2 Calibration and the executed positive control

Phase 1 calibrates the primary Pythia checkpoint before the cross-family comparison. The no-gold floor is 0.095, the gold-only ceiling is 0.650, and moving the gold document from the middle to the first position changes accuracy by $+0.115$ (95% CI $[0.055, 0.175]$). The key-value control produces a $+0.38$ edge-versus-middle effect, demonstrating that the position-intervention pipeline can recover an imposed positional signal.

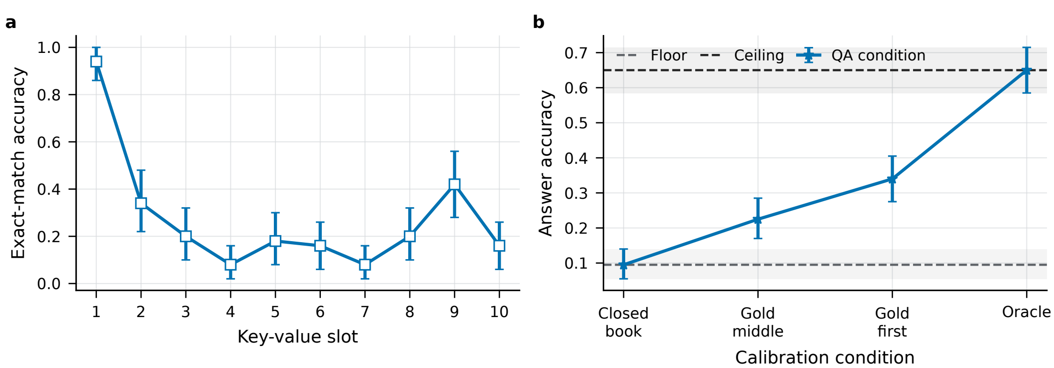
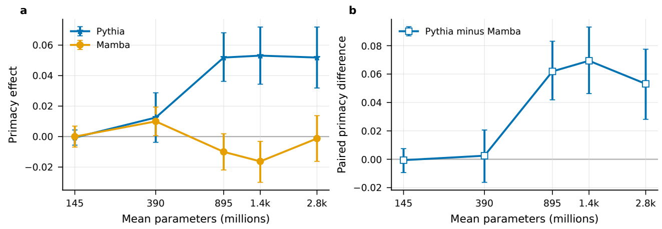


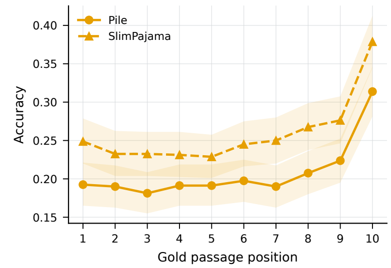
Figure 2: Phase 1 calibration. The left panel shows key-value retrieval by slot. The right panel shows the no-gold floor, middle-document condition, first-document condition, and gold-only ceiling for Pythia-2.8B.

A.3 Scale and corpus scope checks

Figure 3 reports the complete Phase 4 and Phase 5 summaries. The scale trend is inseparable from task capability at the smallest sizes. The corpus comparison changes the accuracy level but does not detect an edge-shape difference under the fixed Mamba-2.8B architecture.



(a) Approximate size pairs



(b) Fixed architecture, different corpora

Figure 3: Additional scope checks. Markers and colors retain the paper-wide encoding: attention-based Pythia models are blue stars, while state-space models are orange and use shape to distinguish checkpoint variants.

A.4 Synthetic retrieval

The Phase 6 forest plot in Figure 4 separates the QA estimates from 2K-token needle retrieval. Open markers denote the synthetic task and filled markers denote QA. Mamba and Mamba-2 saturate the needle task, so the absence of an edge effect for those models is a ceiling result rather than evidence of better middle-context behavior. Only the summary and plots for this phase are committed; raw generations and an environment manifest are not available in the repository snapshot.

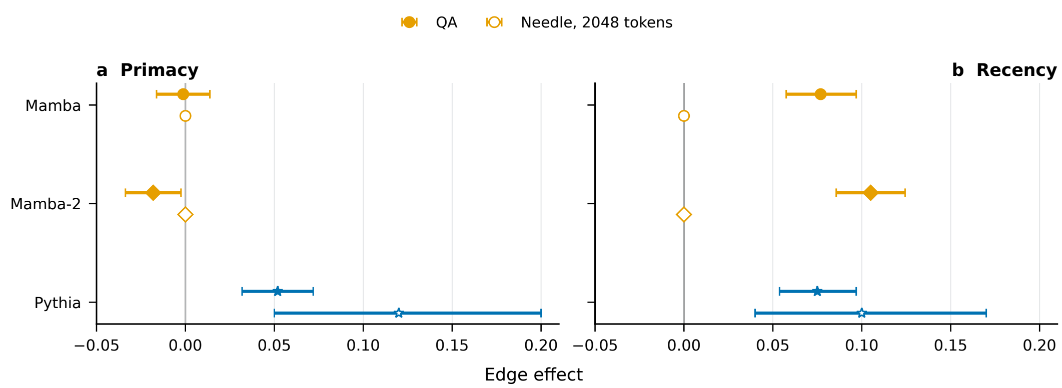


Figure 4: Primacy and recency across the main QA task and 2K-token RULER needle retrieval. Horizontal intervals are 95% question-bootstrap confidence intervals.

A.5 Mechanism-oriented measurements and prompt interventions

Figure 5 combines the valid Phase 7 measurements. The sink statistic and probe results are associations, not interventions. The attempted hybrid attention-sink block is excluded because the model's custom attention path did not honor the hook, so the intervention did not change the computation. The prompt variants jointly alter query position, repetition, and token layout and therefore do not isolate one mechanism.

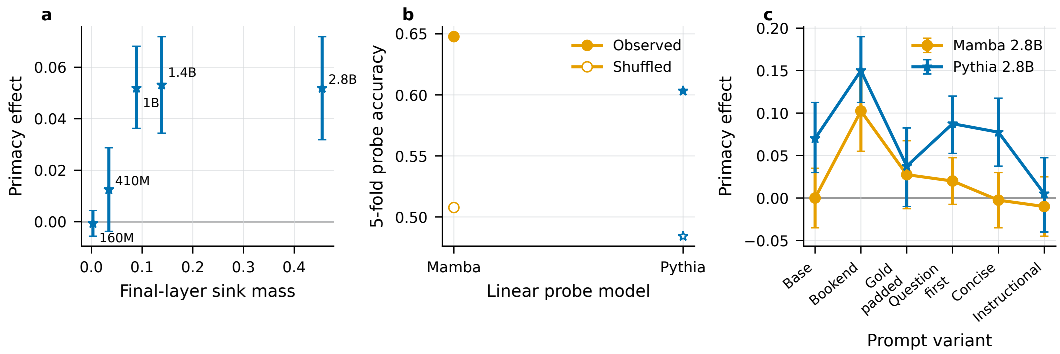


Figure 5: Phase 7 mechanism-oriented evidence. The sink panel compares final-layer initial-token attention share with primacy. The probe panel includes shuffled-label controls. The prompt panel reports edge effects for the executed query and template variants.

A.6 Additional 7B-8B systems

Figure 6 shows the full position curves for the descriptive system comparison. Marker shape and line style distinguish systems without adding colors. These checkpoints differ on several factors beyond the sequence mixer, so the figure is evidence of prevalence rather than a controlled architecture result.

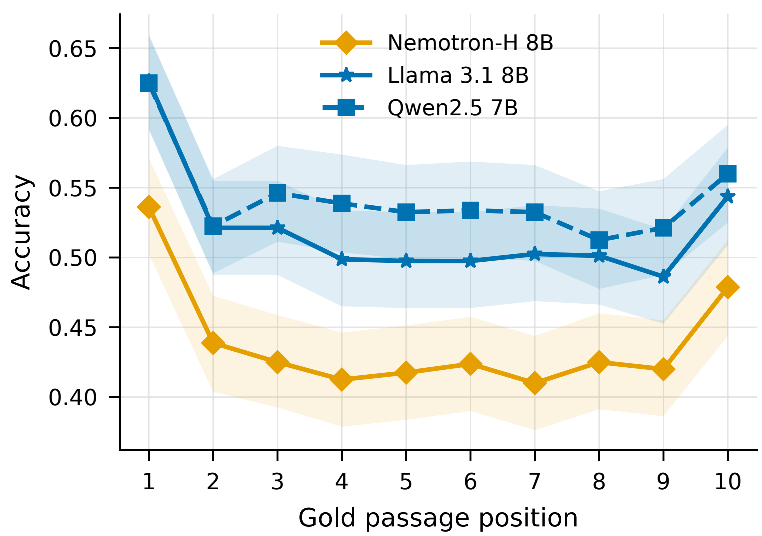


Figure 6: Position curves for Nemotron-H-8B, Llama-3.1-8B, and Qwen2.5-7B on the 800-question exploratory split.

B Integrity and provenance notes

The primary standard sweep yields 9,600 generations per model: 8,000 position prompts, 800 no-gold prompts, and 800 gold-only prompts. The paired bootstrap resamples complete question bundles and preserves prompt and distractor identity across positions. No generation failure or scored null is reported in the primary summaries. Prompt-length checks keep the compared position prompts within one token.

The producing manifest for Phase 3 records uncommitted source changes, so the result is included with an explicit exact-reproduction limitation. A 50-generation blinded audit sample exists, but the human labels were not completed. The negative and distractor-order controls are implemented but have no committed outputs. These missing validations are not counted as passed controls.

Table 4: Execution environments recorded by the phase artifacts.

Phases	Recorded accelerator	Scope
1 and 5	NVIDIA A10G	Calibration and corpus check
2-4	NVIDIA A40	Primary, matched, and scale comparisons
6	NVIDIA T4	Synthetic retrieval
7	NVIDIA L40S and T4	Prompt, sink, and probe studies
8	NVIDIA L40S	Additional 7B-8B systems

B.1 Checkpoint provenance

Table 5 records the immutable revisions for the checkpoints that define the primary, matched, corpus, and additional-system comparisons. The complete scale-suite registry appears in `src/mixing_matters/models.py`.

Table 5: Selected checkpoint repositories and pinned revisions.

Paper key	Repository identifier	Revision
Pythia-2.8B	`EleutherAI/pythia-2.8b`	`2a259cdd96a4`
Mamba-2.8B	`state-spaces/mamba-2.8b-hf`	`96c48e0292b6`
Mamba-2 2.7B	`AntonV/mamba2-2.7b-hf`	`ef542707386f`
Pure Mamba-2 8B	`nvidia/mamba2-8b-3t-4k`	`b915550c63ba`
Hybrid Mamba-2 8B	`nvidia/mamba2-hybrid-8b-3t-4k`	`35e8852e2240`
SlimPajama Mamba-2.8B	`state-spaces/mamba-2.8b-slimpj`	`a7bdd41af90c`
Nemotron-H-8B	`nvidia/Nemotron-H-8B-Base-8K`	`94ea861e008c`
Llama-3.1-8B	`meta-llama/Llama-3.1-8B`	`d04e592bb4f6`
Qwen2.5-7B	`Qwen/Qwen2.5-7B`	`d14972939875`

Elapsed time and aggregate compute were not logged consistently enough to report a defensible total. Checkpoint repository identifiers and immutable revisions are recorded in `src/mixing_matters/models.py` and the per-run manifests. Some evaluated converted checkpoint mirrors do not declare complete license metadata in their model cards, so this snapshot does not claim a complete asset-license audit.

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